Who Gets To Do Nothing?

Team #37: Deepak Alagu, Felix Huang, Simran Mallik, Leah Nuffer, Nayara Paudyal, Yuritzy Ramos

Motivation

Leisure is an essential part of daily life, contributing to physical, mental, and social well-being. However, access to leisure is not evenly distributed. Many individuals face constraints from work demands, caregiving responsibilities, and daily obligations that limit their ability to relax. These disparities reflect a broader form of inequality known as **"time poverty,"** where individuals lack discretionary time despite not necessarily being income-poor.

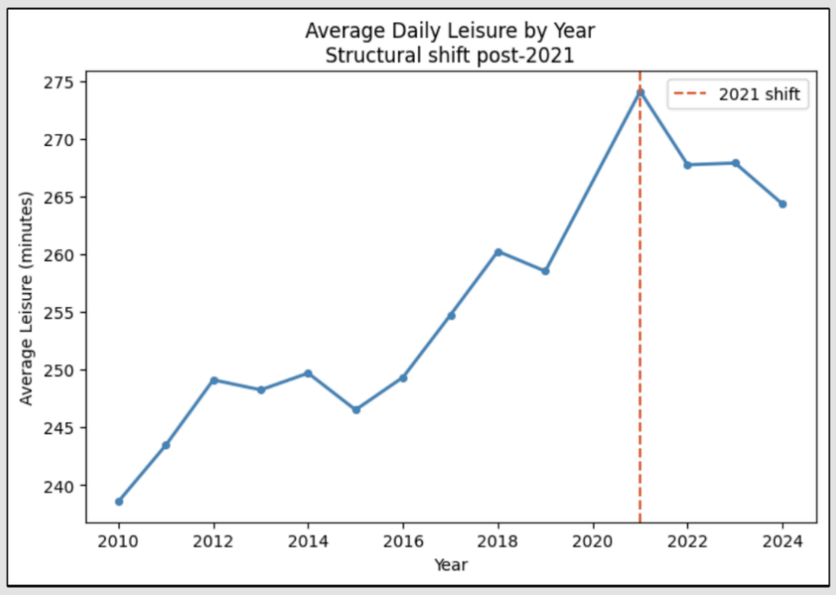
This project aims to understand:

- Who has less time to relax and why
- How schedules and responsibilities shape access to leisure
- How insights can support healthcare professionals and policymakers in designing interventions to improve well-being

Data

We use data from the **American Time Use Survey (ATUS)** from 2010–2024 (excluding 2020), which captures how individuals allocate their time over a 24-hour period. Each observation represents a single-day diary, allowing us to measure daily leisure time and examine how it varies across demographic and socioeconomic groups.

- **Outcome**: daily leisure time (minutes) based on socializing, relaxing, and leisure activities
- **Features** include demographics (age, gender, education), socioeconomic factors (income, employment), and household characteristics (children, spouse presence), along with day-level indicators



Methodology

Our approach combines unsupervised and supervised methods to capture behavioral patterns and key drivers of leisure time.

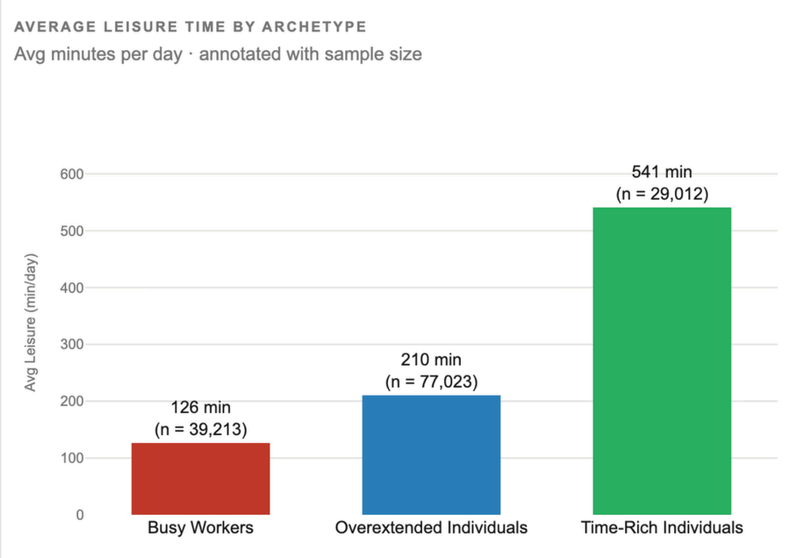
K-Means Clustering:
We apply K-Means clustering on individuals' daily time-use activity proportions to uncover latent behavioral archetypes. We exclude demographic variables to ensure clusters are formed based on how people spend their time, rather than who they are. Using the elbow method, we identify three clusters representing distinct daily schedules. We then analyze these clusters using summary statistics and statistical testing to understand differences in leisure and demographic composition.

Regression:
We use an OLS regression model to predict daily leisure time as a function of demographic and socioeconomic factors. We compare a baseline model with an updated model that includes additional features such as income, work intensity, education, and interaction effects. This allows us to quantify the relative importance of different factors and identify key drivers of leisure inequality, while maintaining interpretability.

Together, these methods provide both interpretable group-level insights and a deeper understanding of how daily constraints shape access to leisure.

Clustering Results

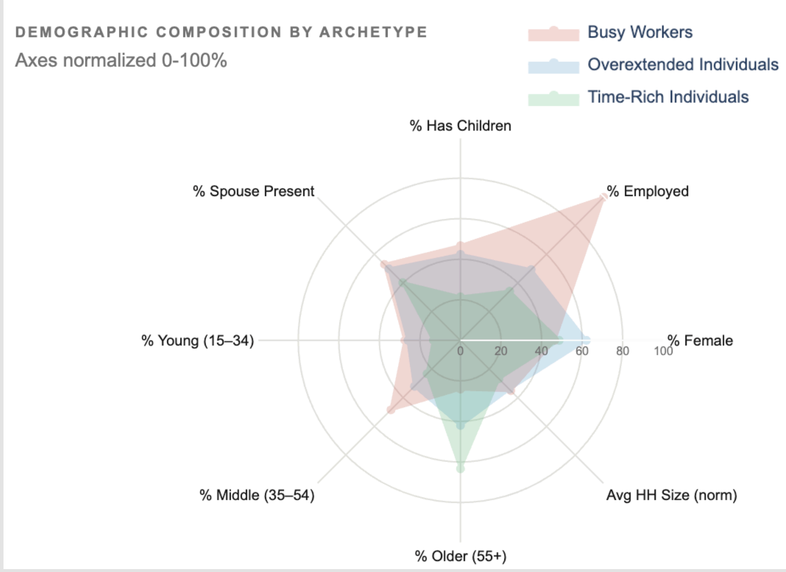
Schedule Archetypes



We discovered 3 schedule archetypes:

- **Busy Workers** have the lowest leisure time averaging 126 minutes per day.
- **Overextended Individuals** spend more time (roughly 1-2 hours more) on household activities than the other groups. They average 210 minutes of leisure time per day and make up the largest amount of observations.
- **Time-Rich Individuals** average 541 minutes per day about 5.5 to 7 hours more per day than the other two groups.

Demographics

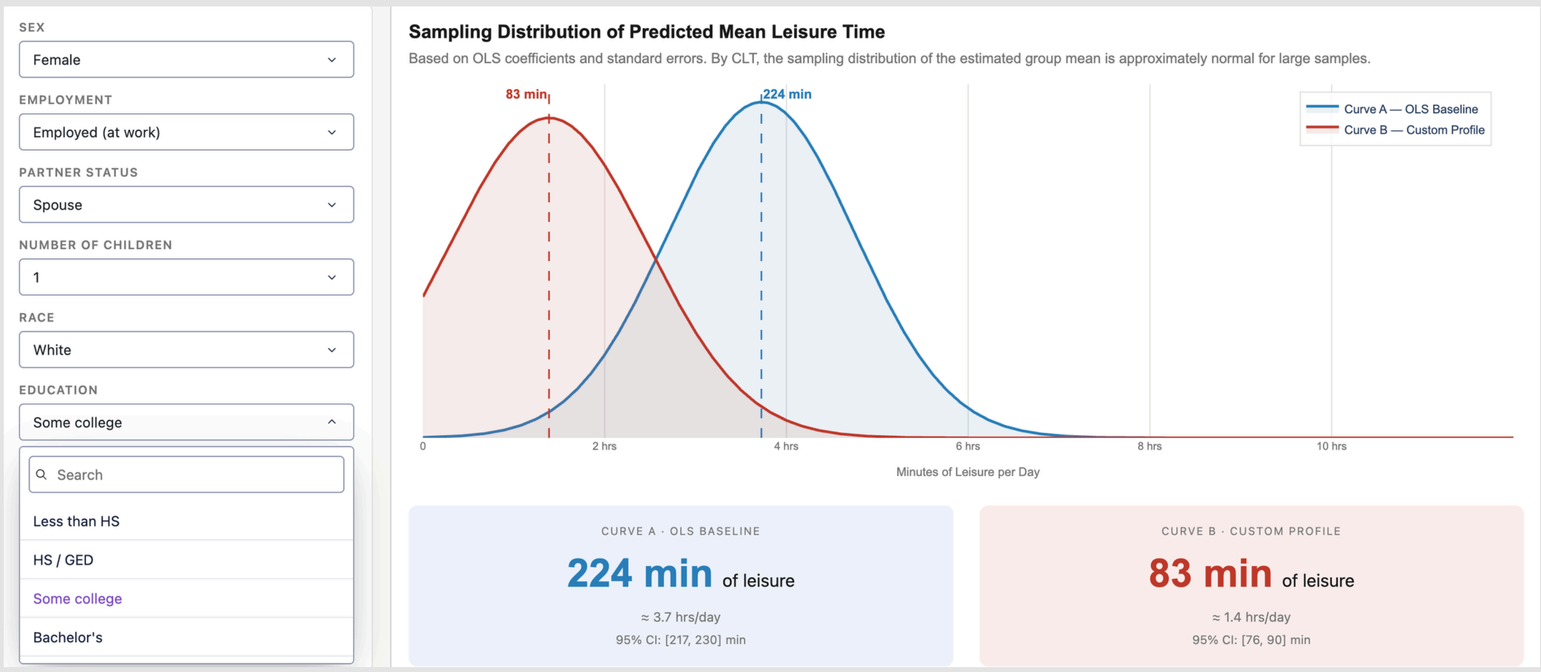


We also looked into the demographics of the 3 cluster groups to observe similarities and differences between them.

- **Busy Workers** are almost entirely employed, and are more likely to be middle-aged (35-54) with a spouse present.
- Around half of **Overextended Individuals** are employed, 62.1% are female, and 42.1% have children.
- **Time-Rich Individuals** are primarily older (55+) and largely not employed.

Regression Results

The baseline OLS model explains 23.5% of the variation in leisure time ($R^2 = 0.2345$), improving to 25.2% ($R^2 = 0.2520$) in the full model, with minimal overfitting.
Employment status is the strongest predictor, with non-employed individuals having significantly more leisure time. We also observe a clear **gender gap**, with women having ~60 minutes less leisure per day.
Other key factors include number of children, work intensity, and education, which are negatively associated with leisure.



The above image shows the leisure time our model predicted for a 29 year-old, white female who is employed, has a spouse , has 1 child, some college education, and weekly earings of $2900 - no part time or multiple jobs reported - on a Monday (non-holiday)

Conclusion

While our clustering and regression analysis found that certain groups tend to have less leisure time, we recognize that **variation still exists within these groups and exogenous variables (outside of demographics) can also effect access to leisure**.Secondly, ATUS data captures only a single day per respondent, providing a snapshot of each individual's time use. Thus the cluster analysis remains **sensitive to day-to-day variability** .

In spite of these limitations, our interactive dashboard **allows users to explore both patterns and predictions directly**. The clustering visualizations highlight differences across schedule archetypes, while the prediction tool enables users to compare a custom profile against a baseline and view predicted leisure time and its associated uncertainty. The dashboard makes model outputs more accessible and demonstrates how multiple factors combine to shape leisure outcomes.This approach provides a **practical tool for both individuals and policymakers** to better understand and visualize how different characteristics and constraints influence access to leisure.

Our findings imply that efforts to address disparities in leisure time should consider not only demographic differences, but also how constraints in daily schedules (e.g., work demands, caregiving responsibilities) shape time allocation.